Python Virtual Environments

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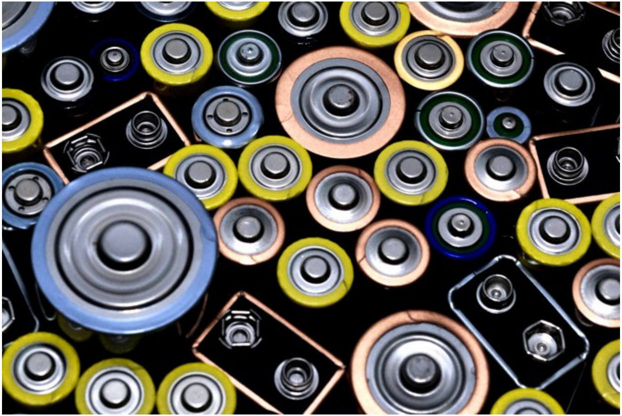
Python Virtual Environments
Outline

Python Modules
Batteries included



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Python Modules

Python Modules and Packages

- python modules are python files containing executable statements and function/class definitions
- python packages group modules together in a directory which also contains a `__init__.py` module
- use them with
  import some_module
- find lots of useful packages from the cheese shop at

  https://pypi.org/

Python Modules
Python Module Search Path

Python modules are searched for in

- the directory containing the script or the current directory
- `PYTHONPATH`, a list of directories (separated by : on Linux/Mac or ; on Windows)
- the installation-dependent default

in a python shell you can check:

```
import sys
print(sys.path)
```

Python Modules

Getting Python Modules

- on Linux - use your distribution package manager such as yum, apt (or speak to your sys-admin to install it for you)
- write your own module and copy it somewhere
- download a module and install iy by hand
  `python ./setup install`
- use
  `pip install some-package`

Python Modules
Installation Location

both `setup.py` and `pip` will

- install into the installation default (you might need admin access)
- `--user`:install into a user writeable location
- `--prefix /some/dir`: specify directory into which to install and set `PYTHONPATH`

Python Environments

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Python Environments

- provide isolated environment into which you can install python modules
- you can have multiple enviornments with different versions

Python Environments
virtualenv

1. install virtualenv using your distribution package manager or `pip`
2. create a virtual environment
   `virtualenv /some/directory`
3. or if you want access to existing package
   `virtualenv /some/directory --system-site-packages`
4. activate the virtual environment
   `source /some/directory/bin/activate`
5. when done run
   `deactivate`

Python Environments
virtualenvwrapper

makes virtualenv more convinient

1. install virtualenvwrapper using your distribution package manager or `pip` (on windows install virtualenvwrapper-powershell)
2. setup a directory for your virtual environments
3. create a virtual environment
   `mkvirtualenv venv`
4. use the virtual environment
   `workon venv`
5. when done
   `deactivate`
6. and finally, remove it
   `rmvirtualenv venv`

Python Environments
virtualenvwrapper

on Linux

```
export WORKON_HOME=~/Envs
# if you installed it using pip you
# need to source the wrapper
source /usr/local/bin/virtualenvwrapper.sh
# make sure the environment directory is created
mkdir $WORKON_HOME
```

on Windows

using powershell

```
$env:WORKON_HOME="~/Envs"
mkdir $env:WORKON_HOME
import-module virtualenvwrapper
```

requirements files record python packages and their versions:

```
package_a
package_b == 0.10.2
package_c > 0.2
```

- install packages from requirements file
  `pip install -r requirements.txt`
- generate requirements file from currently installed packages
  `pip freeze > requirements.txt`

Python Environments
Anaconda

- a python (2 and 3) distribution
- available for Linux, Windows and Macs
- not only python packages but also other libraries such as gdal, etc
- full distribution or bare miniconda

https://www.anaconda.com/

Python Environments
Anaconda

- create a conda environment
  `conda create -n venv`
- or with a specific path
  `conda create -p /some/dir/venv`
- activate environment
  `source activate venv`
- and finally
  `source dactivate`

Python Environments
Anaconda

install packages

- using `pip`
- or

  `conda install package`
- multiple channels such as conda-forge:

  https://conda-forge.org/

  `conda config --add channels new_channel`

Warning

It's easy to get conflicts between packages from different channels and/or the base installation. You can avoid these by installing packages from source using `pip`

- lots of ways of managing conda environments
- similar to `requirements.txt` file you can use an `evironment.yml` file:

```
name:  myenv
channels:
 - conda-forge
dependencies:
 - python=2.7
 - pandas
 - pip:
   - geopandas
```

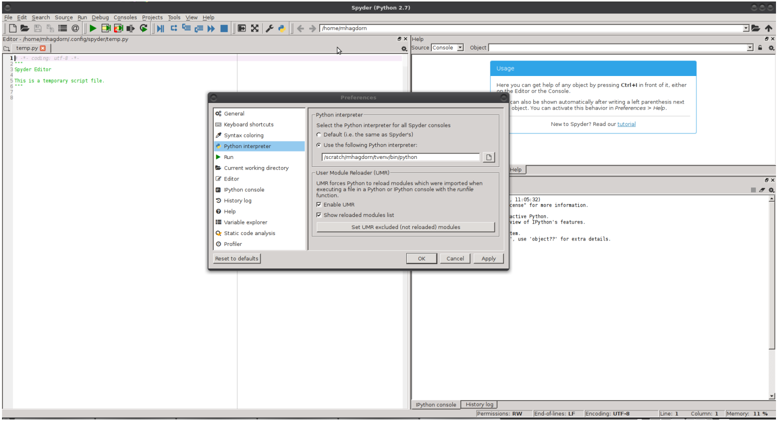
Python Environments
IDEs and Virtual Environments

install IDE into virtual env

- `pip install spyder`
- `conda install spyder`
- in newer versions of spyder you can change the interpreter

Python Environments

IDEs and Virtual Environments



Conclusion

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Recommendations

1. (on Linux) use system packages (or ask your sys-admins)
2. on Windows/Mac use standard conda
3. if you require particular versions use virtual environments
   1. use conda environments if you also need support libraries such as gdal
   2. otherwise use virtualenv/virtualenvwrapper
4. virtual environments are also very useful when you need different versions of your software (production and development version)